

$$\Phi(s) = \frac{KG_1(s)G_2(s)}{1+KG_1(s)G_2(s)}$$



$$D(s) = s^3 + 2.24375s^2 + 0.214375s + 0.01715K$$



Routh condition: $0 < K < 28.046875$



$$K_v = 0.08K$$



$$e_{ss} = \frac{R}{0.08K} \text{ for ramp input}$$